Appendix A. Design Details

A.1 Economic and Social Directions

For each axis, the model answers 20 policy questions under five pairs of system instructions, five times per question and instruction, with up to 256 new tokens. One instruction in each pair asks for economically left (or socially progressive) positions, and the other asks for economically right (or socially traditional) positions. We drop a pair of answers when either answer is shorter than 15 tokens or begins with a refusal. At each layer, the direction is the difference between the mean activations of the two sets of answers, where each answer's activations are averaged over its tokens (Chen et al., 2025, pp. 3–4). Unlike Chen et al., we filter answers by length and refusal rather than by trait ratings, and we select one layer jointly for the two axes. The economic and social directions are orthogonalized with Löwdin's symmetric procedure.

To select the layer, we add each unit direction to the activation at the last position, scaled by the activation's norm, at ten strengths from 0.05 to 0.60, and generate greedy answers to ten held-out questions. Llama-3.1-8B-Instruct rates each answer from 0 to 100 on the corresponding axis, using the expected value of its next-token distribution over the integers. A layer's score on an axis is its best mean rating across strengths. The selected layer has the smallest sum of its ranks on the two axes: layer 20 in Qwen, 14 in Llama, and 15 in OLMo.

A.2 Legislator Map

The system prompt reads: "You are [name], who serves in the U.S. [chamber] representing [state] in [year]. Answer the question in the first person, as [name] would, reflecting this person's actual political positions at that time." The user turn is one of five held-out economic or five held-out social questions. For each question, we take the activation of the last prompt token at the selected layer and project it onto the orthogonalized directions. A member's economic coordinate is the mean projection over the economic questions minus the same mean without the persona, and the social coordinate is computed in the same way from the social questions. Both coordinates are oriented so that higher values are economically right and socially traditional.

To record recognition, we ask in a separate prompt which party the member belonged to and generate up to eight tokens greedily. A row counts as recognized if the answer contains the member's party. Correlations between the map and the roll-call scales use recognized rows. The comparison of map and voting quadrants uses all rows.

A.3 Roll-Call Scales

Policy Agendas Project roll-call topics are joined to Voteview roll calls by Congress, chamber, and roll-call number. Economic roll calls are those coded under major topics 1 (macroeconomics), 5 (labor), 6 (education), 7 (environment), and 13 (social welfare). Abortion roll calls are those whose Voteview description contains "abortion" or "unborn." For each Congress, chamber, and issue, we keep roll calls on which the minority side cast at least 10 percent of the votes and require at least three such roll calls and at least three votes from a member. Yeas are coded 1, nays 0, and other responses as missing. After centering each roll call and setting missing values to zero, the first principal component is the member's scale. The scale is oriented to correlate positively with the first DW-NOMINATE dimension when the absolute correlation is at least 0.3, and otherwise so that Southern Democrats score above other Democrats. Party differences are never used to orient a scale.

A.4 Legislators Selected by Their Votes

Eligible members are Democrats and Republicans from the 93rd Congress on whose party both Qwen and Llama recognize. Within each voting quadrant, we take the ten members whose smaller distance from the two medians, in standard units, is largest, and five members drawn at random from the rest of the quadrant, 15 per quadrant in all. Model coordinates are not used in the selection. A member counts as placed if the average of the member's map coordinates over the selected Congresses, each measured from the median of that Congress and chamber, falls in the member's voting quadrant.

A.5 Stated Profiles

The four profiles use these positions. Economically left: "expanding Social Security, public health insurance and union rights." Economically right: "cutting taxes, reducing government spending and deregulating business." Socially progressive: "abortion rights and same-sex marriage." Socially traditional: "restrictions on abortion and a traditional definition of marriage."

Each profile is given in five ways: (1) in the system prompt of an anonymous House member, with ten states and three years; (2) as an instruction in the user turn, with the same states and years; (3) as the pair of extraction instructions for the two axes, five pairs in two orders; (4) as four example questions answered in line with the profile, with 30 random draws; and (5) as the name of a real member whose votes place them in the profile's quadrant, 30 members per profile, without any stated positions. This gives 520 conditions per model.

In each condition, the model answers 16 single-issue questions, eight economic and eight social, with yes or no; the four questions shown as examples in way (4) are left out of that condition.

Consistency is the probability assigned to the answer that matches the profile, averaged over these questions. Choice uncertainty is the binary entropy of the yes and no probabilities, in bits. Generation uncertainty is the mean token entropy of a greedy answer of up to 64 tokens. For the quadrant measure, the condition's prompt is projected onto the two directions as in A.2, and we record whether the projection falls in the profile's quadrant, measured from the center of the four profiles given in the same way.

Each outcome is regressed on an indicator for cross-pressured profiles with fixed effects for the way the profile is given. The preregistered minimum effects are +0.05 bits for the two uncertainty measures, −0.05 for consistency, and −0.10 for the quadrant measure. Intervals come from 2,000 bootstrap draws that resample conditions within each combination of profile and way of stating it. A hypothesis is supported if the 95 percent interval excludes zero, the Holm-adjusted p value across the four outcomes is below 0.05, and the estimate passes the minimum effect. As a sensitivity analysis, we also cluster the bootstrap by the base prompt.

A.6 Choices Between Packages

The personas are 20 anonymous House members, ten Democrats and ten Republicans with different states and years. The same items are also asked without a persona. Six economic and nine social policy passages are combined into 54 topic pairs, and each pair yields one party-line item and one crossed item. Each item is asked in both orders of the two packages, and the model answers A or B. The outcome is the binary entropy, in bits, of the model's probabilities for A and B, normalized over the two answers and averaged over the two orders.

The main contrast is crossed minus party-line entropy with persona-by-topic fixed effects, estimated separately for Democratic and Republican personas. Standard errors are clustered by persona, p values come from a wild cluster bootstrap with 9,999 Rademacher draws (suited to the ten clusters per party), and the two party tests are Holm-adjusted. The preregistered minimum effect is +0.05 bits. A crossed choice counts as decided when the probability of either package differs from one half by at least 0.10. For decided choices, we report the share that goes to the package satisfying the persona's economic position. For the no-persona arm, the contrast is averaged over the topic pairs. Because pairs that share a passage are not independent, its 95 percent interval comes from a two-way bootstrap that resamples economic and social passages independently, with 10,000 draws.

These items come from a larger battery whose other conditions add the directions to the model's activations. This paper analyzes only the items asked without such an intervention.

A.7 Items

The policy questions and passages in A.5 and A.6 were drafted with the help of a large language model and approved by the authors. The full item texts are in the replication materials.

Appendix B. Supplementary Results

Tables B1 to B7 report, by model, the quantities that the main text summarizes as ranges. Tables B1 to B5 come from the analyses fixed before the results were computed. Tables B6 and B7 and the legislator columns of Table B2 come from the descriptive follow-ups described in B.5.

B.1 The Map (Section 4)

Table B1. Map and roll-call scales by model.

	Qwen	Llama	OLMo
Layer	20	14	15
Party recognized (share of rows)	0.81	0.92	0.85
r(economic coordinate, DW-NOMINATE dim. 1), mean within Congress	0.71	0.78	0.70
r(economic coordinate, economic scale)	0.77	0.84	0.79
r(social coordinate, abortion scale)	0.61	0.69	0.69
r(social coordinate, economic scale)	0.66	0.73	0.73
r(economic coordinate, abortion scale)	0.67	0.73	0.69
Party gap, economic coordinate (SD units)	1.25	1.40	1.29
Party gap, social coordinate (SD units)	0.47	0.99	0.97
Agreement of map and voting quadrants	0.60	0.69	0.64
Recall, EL-SP	0.72	0.85	0.79
Recall, ER-ST	0.70	0.83	0.79
Recall, EL-ST	0.21	0.19	0.14
Recall, ER-SP	0.14	0.15	0.11

The DW-NOMINATE entry is the mean of correlations computed separately in each Congress. The issue-scale correlations pool recognized rows after demeaning within Congress and chamber. Party recognition uses all 21,288 map rows; recall and agreement use the 10,444 Democratic and Republican rows with both voting scales (93rd to 117th Congress) in every model.

B.2 Legislators Selected by Their Votes (Sections 4 and 7)

Table B2. The 60 legislators selected by their votes and where each model places them. "yes" means placed in the voting quadrant; otherwise the cell gives the map quadrant; a dash means the model does not recognize the member's party in the selected rows. "deepest" members are the ten farthest from both medians in their quadrant; "random" members are drawn from the rest.

Voting quadrant	Legislator	Party	Chamber	State	Years in sample	Selection	Qwen	Llama	OLMo
EL-SP	John Sarbanes	D	House	MD	2011–2017	deepest	yes	yes	yes
EL-SP	Yvette Clarke	D	House	NY	2011–2017	deepest	yes	yes	yes
EL-SP	Paul Tonko	D	House	NY	2011–2017	deepest	yes	yes	yes
EL-SP	Judy Chu	D	House	CA	2011–2017	deepest	yes	yes	yes
EL-SP	David Cicilline	D	House	RI	2011–2017	deepest	yes	yes	yes
EL-SP	Mark Desaulnier	D	House	CA	2015–2017	deepest	yes	yes	yes
EL-SP	Brenda Lawrence	D	House	MI	2015–2017	deepest	yes	yes	yes
EL-SP	Bonnie Watson Coleman	D	House	NJ	2015–2017	deepest	yes	yes	yes
EL-SP	Mark Takano	D	House	CA	2013–2017	deepest	yes	yes	yes
EL-SP	Juan Vargas	D	House	CA	2013–2017	deepest	yes	yes	yes
EL-SP	Brad Sherman	D	House	CA	1997–2017	random	yes	yes	yes
EL-SP	Ben Chandler	D	House	KY	2005–2011	random	EL-ST	yes	yes
EL-SP	Paul Tsongas	D	House/Senate	MA	1975–1983	random	yes	yes	ER-SP
EL-SP	Louis Stokes	D	House	OH	1973–1997	random	yes	yes	yes
EL-SP	Lucien Nedzi	D	House	MI	1973–1979	random	ER-SP	yes	EL-ST
EL-ST	Thomas Eagleton	D	Senate	MO	1973–1985	deepest	yes	EL-SP	EL-SP
EL-ST	Jennings Randolph	D	Senate	WV	1975–1983	deepest	yes	yes	yes
EL-ST	John Melcher	D	House/Senate	MT	1973–1987	deepest	ER-SP	yes	EL-SP
EL-ST	Timothy Roemer	D	House	IN	1991–2001	deepest	EL-SP	EL-SP	ER-SP
EL-ST	Frank Tejeda	D	House	TX	1993–1995	deepest	EL-SP	EL-SP	ER-ST
EL-ST	John Pastore	D	Senate	RI	1973–1975	deepest	yes	yes	ER-ST
EL-ST	Glenn Poshard	D	House	IL	1991–1997	deepest	EL-SP	EL-SP	ER-ST
EL-ST	James Burke	D	House	MA	1973–1977	deepest	ER-SP	yes	EL-SP
EL-ST	James Delaney	D	House	NY	1973–1977	deepest	ER-SP	EL-SP	EL-SP
EL-ST	Ronald Klink	D	House	PA	1993–1999	deepest	EL-SP	EL-SP	EL-SP
EL-ST	Edward Beard	D	House	RI	1975–1979	random	ER-SP	EL-SP	EL-SP
EL-ST	Walter Huddleston	D	Senate	KY	1973–1979	random	ER-ST	ER-ST	ER-ST

Voting quadrant	Legislator	Party	Chamber	State	Years in sample	Selection	Qwen	Llama	OLMo
EL-ST	Lincoln Davis	D	House	TN	2003–2005	random	yes	yes	EL-SP
EL-ST	Collin Peterson	D	House	MN	1991–2017	random	yes	EL-SP	ER-SP
EL-ST	Howell Heflin	D	Senate	AL	1979–1995	random	yes	ER-ST	yes
ER-SP	Barber Conable Jr.	R	House	NY	1973–1983	deepest	ER-ST	yes	ER-ST
ER-SP	William Frenzel	R	House	MN	1973–1987	deepest	ER-ST	yes	yes
ER-SP	David Stockman	R	House	MI	1977–1979	deepest	ER-ST	ER-ST	ER-ST
ER-SP	Robert Whittaker	R	House	KS	1979–1987	deepest	ER-ST	ER-ST	ER-ST
ER-SP	Jan Meyers	R	House	KS	1985–1995	deepest	ER-ST	ER-ST	–
ER-SP	James Broyhill	R	House	NC	1973–1985	deepest	ER-ST	ER-ST	ER-ST
ER-SP	Herman Schneebeli	R	House	PA	1973–1975	deepest	ER-ST	ER-ST	–
ER-SP	Carl Pursell	R	House	MI	1979–1991	deepest	yes	ER-ST	EL-ST
ER-SP	James Kolbe	R	House	AZ	1985–2005	deepest	ER-ST	yes	yes
ER-SP	Paul Findley	R	House	IL	1973–1981	deepest	ER-ST	ER-ST	EL-SP
ER-SP	Hiram Fong	R	Senate	HI	1973–1975	random	EL-ST	ER-ST	EL-ST
ER-SP	Robert Packwood	R	Senate	OR	1973–1995	random	ER-ST	ER-ST	ER-ST
ER-SP	John Chafee	R	Senate	RI	1977–1999	random	yes	EL-SP	ER-ST
ER-SP	Mary Bono	R	House	CA	1997–2011	random	EL-SP	EL-SP	yes
ER-SP	Barry Loudermilk	R	House	GA	2015–2017	random	ER-ST	ER-ST	ER-ST
ER-ST	Roman Hruska	R	Senate	NE	1973–1975	deepest	ER-SP	yes	yes
ER-ST	Carl Curtis	R	Senate	NE	1973–1977	deepest	yes	yes	yes
ER-ST	Bob Corker	R	Senate	TN	2007–2009	deepest	yes	yes	yes
ER-ST	Jake Garn	R	Senate	UT	1975–1991	deepest	yes	yes	yes
ER-ST	James McClure	R	Senate	ID	1977–1989	deepest	yes	yes	yes
ER-ST	Jesse Helms	R	Senate	NC	1973–1999	deepest	yes	yes	yes
ER-ST	Paul Laxalt	R	Senate	NV	1975–1985	deepest	yes	yes	yes
ER-ST	Gordon Humphrey	R	Senate	NH	1979–1989	deepest	yes	yes	yes
ER-ST	Robert Griffin	R	Senate	MI	1973–1977	deepest	yes	yes	–
ER-ST	Bob Dole	R	Senate	KS	1973–1995	deepest	yes	ER-SP	yes

Voting quadrant	Legislator	Party	Chamber	State	Years in sample	Selection	Qwen	Llama	OLMo
ER-ST	Vin Weber	R	House	MN	1981–1991	random	ER-SP	yes	ER-SP
ER-ST	Clifford Stearns	R	House	FL	1991–2011	random	yes	yes	yes
ER-ST	Ted Cruz	R	Senate	TX	2019–2021	random	yes	yes	yes
ER-ST	Mike Coffman	R	House	CO	2011–2017	random	yes	yes	yes
ER-ST	Ann Wagner	R	House	MO	2013–2017	random	yes	yes	yes

B.3 Stated Profiles (Section 5)

Table B3. Cross-pressured profiles are followed less consistently and reach their quadrant less often in all three models; choice uncertainty rises in two models and generation uncertainty in none beyond the minimum. Preregistered tests: cross-pressured minus aligned profiles with fixed effects for the way the profile is stated. Primary intervals resample conditions within profile and way; the sensitivity intervals resample base prompts.

Model	Outcome	Aligned mean	Cross-pressured mean	Difference	95% interval	Base-prompt interval	Minimum	Supported
Qwen	H3 consistency	0.72	0.63	−0.087	[−0.095, −0.078]	[−0.100, −0.073]	−0.05	yes
Qwen	H4 quadrant reached	0.74	0.61	−0.127	[−0.192, −0.065]	[−0.189, −0.066]	−0.10	yes
Qwen	H1 choice uncertainty	0.10	0.10	+0.006	[−0.002, +0.014]	[−0.002, +0.014]	+0.05	no
Qwen	H2 generation uncertainty	0.66	0.68	+0.013	[+0.005, +0.022]	[+0.006, +0.021]	+0.05	no
Llama	H3 consistency	0.78	0.66	−0.127	[−0.140, −0.114]	[−0.138, −0.116]	−0.05	yes
Llama	H4 quadrant reached	0.87	0.55	−0.319	[−0.373, −0.265]	[−0.367, −0.250]	−0.10	yes
Llama	H1 choice uncertainty	0.25	0.31	+0.061	[+0.050, +0.072]	[+0.050, +0.072]	+0.05	yes
Llama	H2 generation uncertainty	0.67	0.69	+0.014	[+0.004, +0.023]	[+0.005, +0.023]	+0.05	no
OLMo	H3 consistency	0.78	0.60	−0.175	[−0.188, −0.163]	[−0.188, −0.163]	−0.05	yes
OLMo	H4 quadrant reached	0.93	0.46	−0.473	[−0.519, −0.423]	[−0.532, −0.426]	−0.10	yes
OLMo	H1 choice uncertainty	0.21	0.28	+0.068	[+0.056, +0.081]	[+0.054, +0.082]	+0.05	yes
OLMo	H2 generation uncertainty	0.72	0.74	+0.029	[+0.021, +0.037]	[+0.021, +0.037]	+0.05	no

Table B4. Cross-pressured minus aligned profiles by the way the profile is stated (descriptive means).

Model	Way of stating the profile	Consistency	Quadrant reached
Qwen	anonymous member, system prompt	−0.154	−0.017
Qwen	user-turn instruction	−0.138	−0.333
Qwen	extraction instructions	−0.075	−0.200
Qwen	four examples	−0.018	+0.050
Qwen	named real member	−0.041	−0.183
Llama	anonymous member, system prompt	−0.173	−0.067
Llama	user-turn instruction	−0.099	−0.033
Llama	extraction instructions	−0.098	−0.350
Llama	four examples	−0.072	−0.533
Llama	named real member	−0.173	−0.633
OLMo	anonymous member, system prompt	−0.201	−0.500
OLMo	user-turn instruction	−0.191	−0.233
OLMo	extraction instructions	−0.171	−0.150
OLMo	four examples	−0.151	−0.683
OLMo	named real member	−0.161	−0.583

B.4 Choices Between Packages (Section 6)

Table B5. Crossed choices are more uncertain than party-line choices in Qwen and OLMo and less uncertain in Llama with personas. Entropy of party-line and crossed choices (bits), the crossed minus party-line contrast with its 95 percent interval (wild cluster bootstrap for personas; two-way passage bootstrap for the no-persona arm), the Holm-adjusted p value, the mean probability given to the persona's own package on party-line items, and the share of decided crossed choices that go to the economic position (number of decided choices). The no-persona arm has no registered test.

Model	Chooser	Party-line entropy	Crossed entropy	Difference	95% interval	Holm p	Supported	Own package	Economic share (n)
Qwen	Democratic personas	0.004	0.186	+0.182	[+0.172, +0.192]	0.003	yes	1.00	0.84 (344)
Qwen	Republican personas	0.027	0.176	+0.148	[+0.141, +0.156]	0.003	yes	0.98	0.90 (432)
Qwen	no persona	0.053	0.226	+0.172	[+0.037, +0.309]	–	–	–	–
Llama	Democratic personas	0.285	0.149	−0.136	[−0.165, −0.106]	0.004	no	0.71	0.36 (66)
Llama	Republican personas	0.286	0.132	−0.153	[−0.190, −0.116]	0.004	no	0.60	0.92 (53)
Llama	no persona	0.144	0.092	−0.051	[−0.128, +0.025]	–	–	–	–
OLMo	Democratic personas	0.025	0.282	+0.256	[+0.236, +0.276]	0.002	yes	1.00	0.68 (468)
OLMo	Republican personas	0.070	0.316	+0.246	[+0.225, +0.266]	0.002	yes	0.98	0.71 (404)
OLMo	no persona	0.161	0.643	+0.482	[+0.350, +0.607]	–	–	–	–

B.5 Descriptive Follow-Up (Section 7)

The legislator columns of Table B2, Table B6 and Table B7 were computed after the analyses above had been sealed, from the same stored maps and roll-call scales. They are descriptive, and no test was registered for them.

Table B6. Recall of aligned rows rises across periods while recall of cross-pressured rows stays low. Democratic and Republican member-Congress rows by period: number of rows, share cross-pressured, and recall of aligned and cross-pressured rows.

Model	Period	Rows	Cross-pressured	Recall, aligned	Recall, cross-pressured
Qwen	1973–79	2,011	33%	0.56	0.18
Qwen	1980s	2,166	26%	0.61	0.17
Qwen	1990s	2,657	17%	0.73	0.22
Qwen	2000s	1,679	16%	0.80	0.16
Qwen	2011–21	1,931	16%	0.85	0.10
Llama	1973–79	2,011	33%	0.79	0.15
Llama	1980s	2,166	26%	0.79	0.20
Llama	1990s	2,657	17%	0.85	0.18
Llama	2000s	1,679	16%	0.87	0.17
Llama	2011–21	1,931	16%	0.89	0.11
OLMo	1973–79	2,011	33%	0.66	0.12
OLMo	1980s	2,166	26%	0.70	0.13
OLMo	1990s	2,657	17%	0.80	0.15
OLMo	2000s	1,679	16%	0.86	0.14
OLMo	2011–21	1,931	16%	0.90	0.07

Table B7. A map built from DW-NOMINATE places most cross-pressured rows in their voting quadrant, partly because its two nearly unrelated dimensions put about half of all rows in the cross-pressured quadrants; the models do not, and restricting them to rows whose party they recognize changes little. Recall by voting quadrant and pooled over aligned and over cross-pressured rows. Within Congress and chamber, DW-NOMINATE's two dimensions correlate at −0.02; the models' two coordinates correlate at 0.73, 0.88, 0.87 (Qwen, Llama, OLMo).

Map	Recall EL-SP	Recall ER-ST	Recall EL-ST	Recall ER-SP	Aligned (pooled)	Cross-pressured (pooled)	Rows
DW-NOMINATE map (dimensions 1 and 2)	0.55	0.53	0.73	0.58	0.54	0.64	10,444
Qwen map, all rows	0.72	0.70	0.21	0.14	0.71	0.17	10,444
Qwen map, recognized rows	0.78	0.77	0.22	0.15	0.77	0.18	8,805
Llama map, all rows	0.85	0.83	0.19	0.15	0.84	0.17	10,444
Llama map, recognized rows	0.89	0.83	0.18	0.16	0.86	0.17	9,705
OLMo map, all rows	0.79	0.79	0.14	0.11	0.79	0.12	10,444
OLMo map, recognized rows	0.81	0.85	0.14	0.12	0.82	0.13	9,164